Constantin Alexander Auga

Potsdam, Germany
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Summary

I am a graduate ML researcher working on computer vision, multimodal learning, and predictive representation learning. My research spans multimodal model evaluation, generative modeling, and self-supervised learning, focusing on how models represent information and how those representations support perception, prediction, and world understanding. I am particularly interested in multimodal video learning and predictive models of dynamic environments.

Education

Sydney, AU — **University of Technology Sydney** — **Feb 2027 – Jun 2027**

- Incoming exchange student

Potsdam, DE — **University of Potsdam, Hasso Plattner Institute** — **Oct 2024 – Jun 2027**

- M.Sc. IT Systems Engineering - Grade: **1.1** (1.0 best)
- Thesis: Hearing Beyond the Frame: Investigating multimodal predictive learning for video understanding and action anticipation.

Hamburg, DE — **University of Hamburg** — **Oct 2021 – Sep 2024**

- B.Sc. Computer Science - Final Grade: 2.0 (top 26% of the cohort)

Research Experience

Research Assistant — **Max Delbrück Center** — **Sep 2025 – Present**

- Researched continuous neural representations and convolutional architectures for scalable 2D/3D data, focusing on robustness across resolution changes.
- Designed and evaluated custom metrics and maintained a reproducible research codebase.

Research Assistant — **German Aerospace Center** — **Oct 2023 – Sep 2024**

- Contributed to feasibility studies for hydrogen-powered aviation via experimental data analysis and ML-based leakage localization.
- Built interfaces and data pipelines enabling real-time sensor data ingestion into research databases.
- Developed a real-time heatmap visualization tool, improving experiment monitoring and analysis.

Research, Publications & Projects

TargetSEC (**ICASSP'27** under review)
Constantin Alexander Auga

- Achieved **state-of-the-art** non-duration speech emotion conversion in the arousal on MSP-Podcast by developing a plug-and-play Latent Diffusion Model pipeline.
- Engineered a generative audio architecture, specifically conditioned on arousal within a compact style latent space to improve in-the-wild conversion naturalness.

Beyond Accuracy: Benchmarking Cross-Task Consistency in Unified Multimodal Models (**WACV'27** accepted)
Weixing Wang*, Liudvikas Zekas*, Anton Hackl*, **Constantin Alexander Auga***, et al. (* Equal contribution)

- Quantified representational consistency across 8+ multimodal models (including **Gemini** and **GPT-5**), exposing coherent hallucination failure modes that standard metrics fail to detect.
- Introduced scene-graph-grounded metrics to decouple representational consistency from factual accuracy, enabling granular evaluation of cross-task visual semantic stability.

Jaguar Re-Identification Challenge | Python (PyTorch)

- **Won 1st place** in a Kaggle competition, ranking **#1 out of 350+ teams** with **370+ participants**.
- Built a hybrid DINOv3-based re-identification model for fine-grained visual matching.
- Improved leaderboard performance by 25% in comparison to the baseline model.

Volunteering

Student Representative **University of Potsdam, Hasso Plattner Institute** **Nov 2025 - Present**

- **Evaluated** candidates for two open professorship positions as an appointed member of the official selection committee
- **Facilitated** discussions between faculty and students to assess candidate qualifications and teaching capabilities.
- **Represented** student interests and formally presented candidate recommendations to the university administration.

Youth Volleyball Coach **Altonaer Turn- und Sportverein** **2020 - 2024**

- **Served** as a head coach for youth teams at Hamburg's largest volleyball club.
- **Taught** fundamental techniques and team strategies to **squads of 8-10 children** aged 12 to 16.
- **Mentored** and led youth teams through regular competitive league games and tournaments.

Technical Skills

- **Programming Languages**: Python, C++, C#, Java, JavaScript, SQL
- **Technologies**: PyTorch, HuggingFace, Git, NumPy, Pandas, Scikit-learn, Linux, ROS

Miscellaneous

- **Languages:** German (native), English (TOEFL iBT 103 / CEFR C1), Portuguese (basic), Spanish (basic), Georgian (basic)
- **Hobbies and Interests:** Volleyball & Team Sports, Traveling